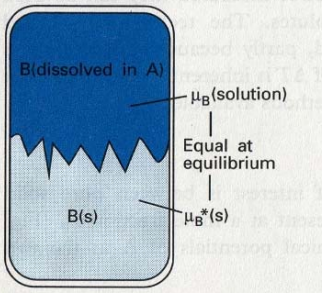
Solubility in Ideal Systems: Integration within the CRYSS Framework

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Introduction

The following discussion draws upon principles articulated in *Physical Chemistry* (5th Edition) by P.W. Atkins (pp. 226-227). In the highlighted excerpt, Atkins addresses the inherent limitations of the Ideal Approximation, specifically noting that such a framework cannot provide information regarding the absolute solubility of a compound or mixture. In real-world systems—even those approaching ideal behaviour—predicting absolute solubility in solvents such as ethanol or acetone via the Schroeder van Laar relationship (eqn 20) is not possible. However, this limitation does not negate the significant utility of analysing chemical systems through the lens of the Ideal Approximation.



7.17 The heterogeneous equilibrium involved in the calculation of the solubility is between pure solid B and B in the mixture.

Solubility

Although it is not strictly a colligative property, the solubility of a solute may be estimated by the same techniques as we have been using. If a solid solute is left in contact with a solvent, it dissolves until the solution is saturated. Saturation is a state of equilibrium, with the undissolved solute in equilibrium with the dissolved solute. Therefore, in a saturated solution the chemical potential of the pure solid solute, $\mu_B^*(s)$, and the chemical potential of B in solution, μ_B, are equal (Fig. 7.17). Because the latter is

$$\mu_B = \mu_B^*(l) + RT \ln x_B$$

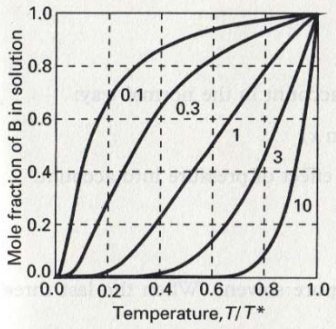
we can write

$$\mu_B^*(s) = \mu_B^*(l) + RT \ln x_B$$

This expression is the same as the starting equation of the last section, except that the quantities refer to the solute B, not the solvent A.

The starting point is the same but the aim is different. In the present case we want to find the mole fraction of B in solution at equilibrium when the temperature is T. Therefore, we start by rearranging the last equation to

$$\ln x_B = \frac{\mu_B^*(s) - \mu_B^*(l)}{RT} = -\frac{\Delta_{fus}G}{RT}$$
$$= -\frac{\Delta_{fus}H}{RT} + \frac{\Delta_{fus}S}{R}$$

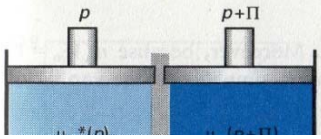


7.18 The variation of solubility (the mole fraction of solute B in a saturated solution) with temperature (T^* is the freezing temperature of the solute). Individual curves are labelled with the value of $\Delta_{fus}H/RT^*$.

At the melting point of the solute T^* we know that $\Delta_{fus}G = 0$, and thus $\Delta_{fus}G/RT^* = 0$ too; consequently, the term $\Delta_{fus}G/RT^*$ may be added to the right-hand side. If the enthalpy and entropy of melting are constant over the temperature range of interest, then the entropy terms cancel and, in much the same way as before, we obtain

$$\ln x_B = -\frac{\Delta_{fus}H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right) \qquad (20)^\circ$$

This expression (which is plotted in Fig. 7.18) shows that the solubility of B decreases exponentially as the temperature is lowered from its melting point and that solutes with high melting points and large enthalpies of melting have low solubilities at normal temperatures. However, the detailed content of eqn 20 should not be treated too seriously because it is based on highly questionable approximations, such as the ideality of the solution. One aspect of its approximate character is that it fails to predict that solutes will have different solubilities in different solvents, for no solvent properties appear in the expression.



The Utility of the Ideal Approximation in Enantiomer & Diastereomer Systems

The practical value of the ideal approach is most evident in the analysis of binary melting point phase diagrams for enantiomeric systems and particularly the more common appearance of racemic compounds (1:1 addition compounds); here, the eutectic point allows for the calculation of the maximum theoretical yield obtainable from a partially enantioenriched mixture (e.g., following asymmetric synthesis). Consequently, the analysis of phase diagrams for diastereomeric systems[1] is typically obtained whether performed *via* traditional methods or Differential Scanning Calorimetry (DSC).[2]

The utility of these phase diagrams in enantiomeric and diastereomeric systems arises from the structural similarity of the components. While the Ideal Approximation cannot predict absolute solubility, we can infer that because the chemical functionalities—the solvating moieties—are nearly identical, their differential solubilities across various solvents will typically remain minimal. Essentially, the solvent lacks the structural differentiation required to significantly alter the relative solubility of the components.

Based on the melting binary phase diagram of diastereomeric mixtures, Ács et al. [3] calculated that, based on the knowledge of the eutectic composition of the binary melting phase diagram, it is possible to calculate the resolvability (*F*) of the crystalline precipitating diastereomer of the given racemic compound with the given resolving agent (**Figure 2**). This is a generally applicable method if, after optimization, diastereomeric mixtures have been obtained, and their melting binary phase diagram can be plotted.

$$F = \frac{1 - 2\frac{ee_{EuDia}}{100}}{1 - \frac{ee_{EuDia}}{100}} \quad (2)$$

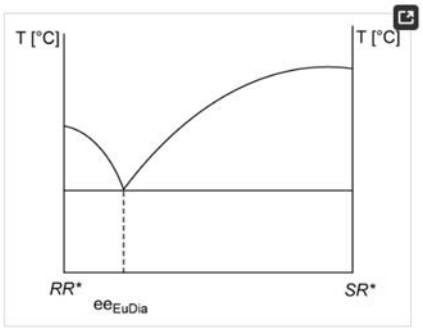


Figure 2. Binary melting phase diagram of a diastereomeric mixture of *RR** and *SR** [3,6] and the possibility of calculating the achievable *F* value using the ee_{Dia} (Equation (2)).

Consequently, when evaluating mixtures, it is reasonable to expect that systems composed of structurally similar components will exhibit behaviour closer to ideality than more disparate, structurally diverse mixtures.

The NODES Application: Deterministic Modelling

The NODES application extends these fundamental principles by generalizing them across multicomponent systems using the Ideal Approximation. Utilizing initial composition data, melting points, and heats of fusion, NODES calculates the idealized purification profile of a system (as shown in Figure 1). This allows for the prediction of impurity migration as a function of **Recovery**.

[1] https://en.wikipedia.org/wiki/Diastereomeric_recrystallization

[2] Pálovics, E.; Madarász, J.; Pokol, G.; Fogassy, E.; Bánhegyi, D.F. Economic Separations of Organic Acidic or Basic Enantiomeric Mixtures—A Protocol Suggestion. *Int. J. Mol. Sci.* **2023**, *24*, 846. https://doi.org/10.3390/ijms24010846

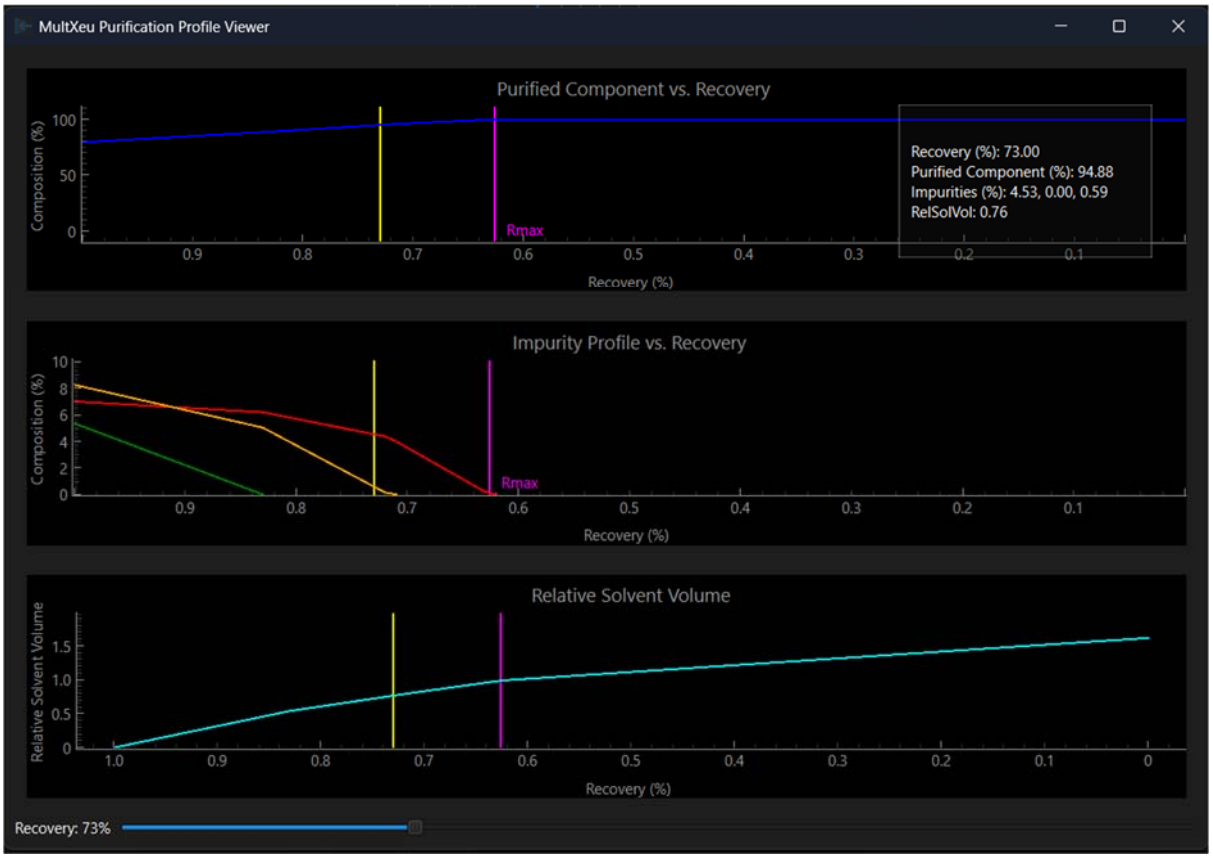


Figure 1: A plot illustrating the NODES application routine, showing the relationship between Recovery and the purification profile.

In this context, **Recovery** is defined as the state of the system following the equilibration of a mixture between the solid and solution phases. In a zero-solvent scenario, the system exhibits 100% Recovery (all material remains in the solid phase). Conversely, if 10% of the material is dissolved, the solid phase retains 90% Recovery. Because we are analysing equilibrium behaviour, the more soluble eutectic component is preferentially partitioned into the solution phase due to its lower melting point—a direct application of binary phase diagram logic.

By applying the Schroeder van Laar equation (and its counterpart for 1:1 addition compounds), we can calculate the solubility of the mixture throughout the "Recovery journey." This enables the determination of the **Relative Solvent Volume** at any arbitrary recovery point. This calculation reveals a critical threshold: there exists a specific volume of solvent associated with the maximum yield (R_{max}) of the primary component. Utilizing more solvent than this threshold results in a loss of yield of pure product to the mother liquor.

The CRYSS Framework: Empirical Refinement

While NODES does not account for absolute solubility or deviations from ideality, it serves as an excellent engine for generating self-consistent datasets with a fundamental link to the underlying equilibrium partitioning of a mixture between solid and solvent phases. Of particular interest is the system behaviour between 100% Recovery and R_{max} (refer to Figure 1). By sampling the system at

various recovery points within this region, we can fit a mathematical model to the observed solubility behaviour. This model constitutes **CRYSS** (Crystallization Screening System).

The strength of CRYSS lies in its independence from specific physical property data; instead, these properties are encoded within the observed purification behaviour. The methodology employed in CRYSS makes use of fitting a thermodynamically relevant model to a real system. The primary feature of CRYSS is that captures the actual solubility behaviour of the specific solvent system by absorbing deviations from ideality into the fitted model.[3] This approach provides a realistic estimation of R_{max} in real-world chemical systems.

Experimental Considerations and Methodology

To ensure the integrity of the model, the experimental protocol must guarantee that the system has reached true thermodynamic equilibrium. Furthermore, sampling must be restricted to the region between 100% and R_{max}, as the loss of degrees of freedom beyond R_{max} precludes accurate modelling. This implies working with relatively small solvent volumes, a technique well known to those who screen enantioenriched systems or classical resolutions, since they benefit from the simplicity of a binary system where the solvent phase in the region 100% - R_{max} is, at equilibrium, the eutectic composition; that is not true in multicomponent systems as is apparent from Figure 1.

Key practical considerations include:

- **Equilibration:** Utilizing thermal cycling to ensure complete phase equilibrium.
- **System Stability:** Ensuring the system is closed and chemically/thermally stable under experimental conditions.
- **Sampling Precision:** Accurate sampling of the mother liquor is critical; contamination by the solid phase must be strictly avoided.

An effective approach involves using closed systems that can be centrifuged after equilibration, followed by sampling via needle through in-line filtration. If the initial composition is known (e.g., by dispensing precise volumes into a reactor array and removing solvent via vacuum to ensure homogeneity), the mass balance can be calculated by assaying the equilibrium mother liquor.[4] This allows for the determination of both the solid composition and the exact Recovery at the point of analysis. Provided this mass balance is accurate, it provides all the necessary data for CRYSS

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[3] The Thermodynamic Solubility Equation: $a_i = \gamma_i . x_i = e^{Q}$; where $Q = \frac{\Delta H_f}{R}\left(\frac{1}{T_f} - \frac{1}{T}\right)$, a_i: is the activity of the solute in the solution, γ_i: is the activity coefficient of the solute in the solvent, x_i: is the mole fraction of the solute in the solution, ΔH_f= is the enthalpy of fusion (melting) of the pure solid solute, T_f= is the melting temperature (in Kelvin) of the pure solid solute, R= is the ideal gas constant, T= is the absolute temperature (in Kelvin) of the system. Thus, in the CRYSS approach we are using real, non-ideal, solubility behaviour of a system we are simply fitting the lumped terms in the exponential; since the measured mass fraction (m_i) is proportional to the mole fraction (i.e. $m_i = \beta_i . x_i$), and therefore, $m_i = e^{Q + \ln(\beta_i) - \ln(\gamma_i)}$, this is how CRYSS is capturing the real system information supplied from analytical measurements.

[4] The recovery may also be calculated in a closed system via the lever rule: $R = \frac{x_{ci} - x_{li}}{x_{si} - x_{li}}$; where with respect to each component in the system x_{ci} relates to the starting composition, x_{li} relates to the liquor composition at the point of sampling, x_{si} relates to the solid composition at the point of sampling. Thus, a reasonable estimate of R may be estimated for each component, and practicality improved by clustering the values and outlier rejection.

calculations. As with any robust experimental protocol, the use of replicates and varying solvent volumes (ensuring all points remain within the 100% – R_{max} range) significantly enhances the predictive power of the model.

This systematic approach is highly compatible with parallel reactor arrays, a technology that is widely available through various levels of commercial sophistication.







